



FEDERAL PUBLIC SERVICE COMMISSION
COMPETITIVE EXAMINATION-2024 FOR RECRUITMENT
TO POSTS IN BS-17 UNDER THE FEDERAL GOVERNMENT
PURE MATHEMATICS

Roll Number

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS = 100

- NOTE: (i)** Attempt **FIVE** questions in all by selecting **TWO** Questions each from **SECTION-A&B** and **ONE** Question from **SECTION-C**. **ALL** questions carry **EQUAL** marks.
- (ii)** All the parts (if any) of each Question must be attempted at one place instead of at different places.
- (iii)** Write Q. No. in the Answer Book in accordance with Q. No. in the Q.Paper.
- (iv)** No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- (v)** Extra attempt of any question or any part of the attempted question will not be considered.
- (vi)** **Use of Calculator is allowed.**

SECTION-A

- Q. No.1.(a)** Let N be a normal subgroup of a group G . If H is a subgroup of G , then prove that $NH = \{nh : n \in N \text{ and } h \in H\}$ is a subgroup of G . (10)
- (b)** If E is an epimorphism from a group G onto a group H then prove that G/K is isomorphic to H , where $K = \text{Ker } E$. (10) (20)
- Q. No.2. (a)** Let R be a ring. If every $x \in R$ satisfies $x^2 = x$ then prove that R is a commutative. (10)
- (b)** For which value(s) of a will the following system have no solution? Exactly one solution? Infinitely many solutions? (10) (20)
- $$\begin{aligned} x + 2y - 3z &= 4 \\ 3x - y + 5z &= 2 \\ 4x + y + (a^2 - 14)z &= a + 2. \end{aligned}$$
- Q. No.3. (a)** Determine a basis for and the dimension of the solution space of the system (10)
- $$\begin{aligned} x - 2z + w &= 0 \\ 3x + y - 5z &= 0 \\ x + 2y - 5w &= 0. \end{aligned}$$
- (b)** Let $v_1 = (1, 1, 1)$, $v_2 = (1, 1, 0)$ and $v_3 = (1, 0, 0)$ be a basis for $\mathbb{R}^3$. Find a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(v_1) = (1, 0)$, $T(v_2) = (2, -1)$ and $T(v_3) = (4, 3)$. (10) (20)

SECTION-B

- Q. No.4. (a)** Evaluate the limit: (10)
- (i)** $\lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos x)^{\tan x}$
- (ii)** $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$
- (b)** State and prove the Mean Value Theorem. (10) (20)
- Q. No.5. (a)** If $w = f(x^2 + y^2)$ then show that $y \left(\frac{\partial w}{\partial x} \right) - x \left(\frac{\partial w}{\partial y} \right) = 0$. (10)
- (b)** Find all the local maxima, local minima and saddle points of the given function $2x^3 + y^2 - 9x^2 - 4y + 12x - 2$. (10) (20)

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Q. No.6. (a) Evaluate the integral $\int_0^{\infty} x^{\frac{3}{2}} ((1+2x))^{-5} dx$ and show that the result is $\frac{9\pi}{384}$, (10)
using **Beta** function.

(b) Find the vertices and foci of the hyperbola (10) (20)
 $25x^2 - 16y^2 + 250x + 32y + 109 = 0$.

SECTION-C

Q. No.7. (a) Verify that $u(x,y) = \cos x \cosh y$ is harmonic function and find a (10)
corresponding analytic function $f(z) = u(x,y) + iv(x,y)$.

(b) Use Residue theorem to evaluate the integral $\int_C \frac{3z^2 + z - 1}{(z^2 - 1)(z - 3)} dz$, where C is (10) (20)
the
circle $|z| = 4$.

Q. No.8. (a) Use the Cauchy's integral formula to evaluate the integral (10) (10)
 $\int_C \frac{z+4}{z^2 + 2z + 5} dz$, where C is the circle $|z+1-i| = 2$.

(b) Find the three cube roots of $\sqrt{3} + i$. (10) (20)
